2026 International Conference on Emerging Smart Computing and Informatics (ESCI)
Pune, India. Mar 11-13, 2026

GreenFederate: A Federated Learning Framework for Energy-Efficient and Privacy-Preserving Model Training on Distributed Edge Devices

Mohammad Abu Kausar
Department of Information Systems, University of Nizwa,
Nizwa, Sultanate of Oman
kausar4u@gmail.com

Mohammad Nasar
Computing and Informatics department, Mazoon College,
Muscat, Sultanate of Oman
nasar31786@gmail.com

S. M. Emdad Hossain
Department of Information Systems, University of Nizwa,
Nizwa, Sultanate of Oman
emdad.hossain@unizwa.edu.om

2026 International Conference on Emerging Smart Computing and Informatics (ESCI) | 979-8-3315-8949-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/ESCI68015.2026.11493260

***Abstract*— Federated Learning (FL) is a collaborative model training paradigm that allows to train models among distributed devices collaboratively with no need of raw data transfer, on one hand protecting users' privacy. But traditional FL is still highly energy-costly and communication-expensive, especially in the context of heterogeneous resource-limited edge settings, with high emission of carbon and far-from-optimal learning dynamics. In this paper, we propose GreenFederate, a comprehensive approach which leverages MARL-based client selection, adaptive sparsification, knowledge distillation followed by hierarchical aggregation and calibrated differential privacy. GreenFederate also includes carbon-aware scheduling that schedules training during low-carbon renewable-energy availability. Comprehensive experimental results over a variety of image and sensor datasets, along with real-device experiments over Raspberry Pi clusters, show that GreenFederate achieves 50.8% energy reduction, 51% cccarbon savings and reduces communication cost by 60-80%, while achieving strong accuracy in non-IID scenarios. The envisioned framework paves a way towards sustainable, privacy-aware edge intelligence.**

Keywords*— *Federated Learning, Edge Computing, Energy Efficiency, Multi-Agent Reinforcement Learning, Differential Privacy, Hierarchical Aggregation, Green AI.

I. INTRODUCTION

The exponential growth of edge devices smartphones, sensors and IoT (Internet of Things) nodes, has given raise to an unprecedented amount of decentralized data. Federated Learning (FL) is an emerging paradigm to support devices learn a model collaboratively without directly transmitting raw data to the central server for training, which can mitigate privacy breach and communication burden [9]. Such advantages notwithstanding, traditional FL bears high energy and carbon cost, mainly because of frequent parameter exchanges, iterated local computations and inefficient client participation strategies [10], [17].

Recent work discovers that FL can be more energy intensive by 2–5× compared with centralized training under the present of device heterogeneity and unreliable-wireless environment [10], [19]. At the same time, gradient inversion and model-leakage attacks illustrate that stronger privacy-preserving mechanisms that adapt beyond naïve parameter sharing are important [9], 15]. These challenges highlight a need for FL systems that not only maximize energy efficiency, but also consider privacy, communication cost and model utility together in order to enable practical edge deployments.

This work proposes GreenFederate, a unified framework that addresses these challenges through a multilayer optimization strategy combining: MARL-based user selection, Adaptive sparsification with knowledge distillation, Hierarchical federated aggregation, Layer-wise calibrated differential privacy and Carbon-aware scheduling.

These components taken jointly constitute an energy-efficient and privacy-preserving FL pipeline tailored for practical resource-limited settings.

Contributions

The main contributions of this work are as follows:

1. **Unified Framework:** We present GreenFederate, a holistic FL architecture that co-optimizes energy consumption, privacy and communication overhead across the heterogeneous edge networks.
2. **Energy-Aware MARL Selection:** We develop a lightweight module for MARL to select clients with respect to the energy level, channel state, learning contribution, and renewable-energy availability which delivers 22% energy cost reduction.
3. **Adaptive Sparsification with Distillation:** We propose a two-phase compression pipeline that reduces messages by up to 80% while at the same time maintaining model accuracy.
4. **Hierarchical Aggregation with Layered DP:** We present a clustering aggregation design with calibrated differential privacy, which achieves 10-15% better accuracy than flat DP.
5. **Carbon-Aware Training Scheduler:** We use renewable-energy forecasts to shift elastic training in time and reduce carbon footprints by 51%.
6. **Extensive Evaluation:** We conduct thorough experiments on five datasets and the real Raspberry Pi devices deployed, showing efficiency, robustness, and scalability.

To our knowledge, GreenFederate is the first FL framework to co-design MARL client selection, adaptive sparsification, distillation, hierarchical DP, and carbon-aware scheduling as one deployable system.

II. RELATED WORK

A. Energy-Efficient FL

There are several proposed methods to ensure active learning in FL is energy efficient. The renewable-energy-aware scheduling based on Lyapunov optimization was introduced to GreenFL [3]. Fed2Tier [5] proposed hierarchical

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aggregation to alleviate global communication overhead. 70 to 90 % reduction on the uplink [8]), and multi-layer resource coordination has been studied for CPU-frequency / batch-size adaptation [6]. Reinforcement learning, such as MARL, is more and more employed for dynamic clients scheduling under limited resources [7], [11]. Theoretical and simulation energy modeling results reveal the trade-offs of computation versus communication cost [10], [12].

B. Privacy-Preserving FL

Differential privacy (DP) remains a widely adopted mechanism for privacy protection in FL, with concentrated and Rényi DP offering tighter bounds [9]. Hierarchical aggregation further enhances privacy by reducing exposure of individual updates [13], [14]. Secure aggregation and homomorphic encryption provide strong guarantees but suffer from high computational overhead [20]. Lightweight DP methods for mobile-health applications are also emerging [15].

C. Integrated Green-Private FL

Several surveys emphasize the intersection between sustainability and privacy in FL [18]. Eco-friendly data selection and avoidance of redundant updates have been studied [17], and industrial IoT research highlights communication-energy-robustness tradeoffs [19]. However, existing works optimize only one or two aspects. GreenFederate is the first to unify energy, privacy, sparsification, MARL, and carbon-aware scheduling into one cohesive framework.

III. System Model and Problem Formulation

A. Network and Device Model

We consider N heterogeneous edge devices participating in FL. Each device i possesses:

- A local dataset $\mathcal{D}_i$, $|\mathcal{D}_i| = d_i$
- Maximum battery capacity $E_i^{\max}$
- Residual energy $E_i(t)$
- Computing frequency f_i
- Transmit power P_i^{tx}

A central server coordinates training rounds via K cluster heads, enabling hierarchical aggregation. Wireless communication occurs over channels with bandwidth B and noise $\mathcal{N}_0$.

B. Energy Consumption Model

The per-round energy consumption of device i is:

$$E_i = E_i^{comp} + E_i^{comm} = \kappa f_i C + \frac{s_i}{r_i} \cdot P_i^{tx}$$

where κ is the processing-energy coefficient, C is the required FLOPs, s_i is the update size, and r_i is the uplink rate.

Carbon footprint is modeled as:

$$CF = \gamma \cdot E_{\text{total}}$$

where γ is grid carbon intensity.

C. Optimization Objective

GreenFederate minimizes the global loss while respecting energy and privacy constraints:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \sum_{i=1}^{N} p_i F_i(\mathbf{w})$$

subject to:

$$E_i(t) \le E_i^{\max}, \sum_i E_i(t) \le B_{\text{total}}, (\epsilon, \delta)\text{-DP}, \text{Accuracy} \ge \theta$$

This formulation explicitly captures the energy, privacy, and accuracy requirements critical for sustainable FL.

IV. GreenFederate Framework Design

This section presents the design of GreenFederate, a unified framework for energy-efficient, privacy-preserving, and carbon-aware federated learning (FL) on distributed edge devices. The framework integrates five synergistic components:

1. MARL-based client selection
2. Adaptive sparsification
3. Knowledge distillation for stable local training
4. Hierarchical aggregation with layered differential privacy (DP)
5. Carbon-aware scheduling for sustainable global rounds

The architectural overview is shown in Fig. 1.

A. System Architecture

GreenFederate utilizes a network hierarchy of devices, cluster heads, and a global server. Devices conduct local training, sparsify their updates and transmit them in the form of compressed gradients to dedicated cluster heads. These nodes carries out intermediate aggregation, they add DP noise and send the aggregate updates to the central server.

This pipeline dramatically reduces the communication, distributes the privacy cost and reduces energy consumption.

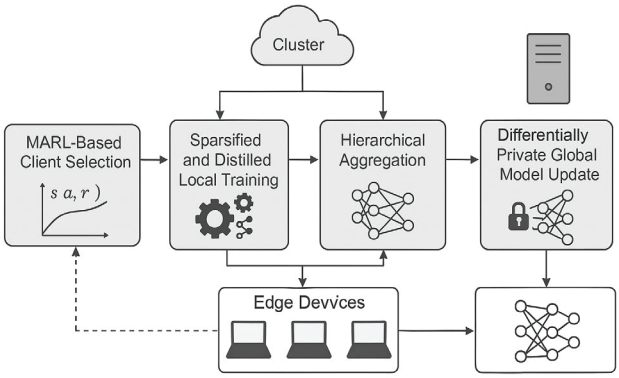


Fig. 1. System Architecture of the GreenFederate Framework.

The architecture overcomes drawbacks noticed in previous hierarchical and energy-aware FL designs [3], [10], [13], [19] by integrating client including, compression, privacy, and sustainability tactics into a unified system.

B. Energy-Aware Client Selection via MARL

Each device hosts a lightweight MARL agent with state:

$$s_i = [E_i, h_i, \Delta acc_i, RE_t]$$

The agent decides whether to participate:

$$a_i \in \{0,1\}$$

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The reward penalizes energy consumption and encourages accuracy gain and renewable-energy alignment.

The MARL agent observes a multi-dimensional state and outputs a binary participation action. Fig. 2 illustrates the client-selection process, including state extraction, policy evaluation, action generation, and reward feedback.

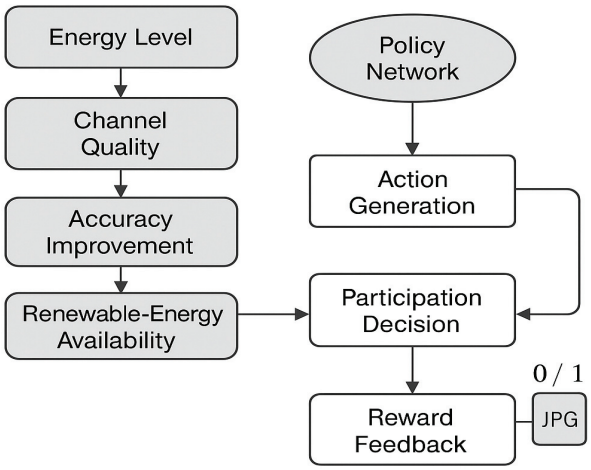


Fig. 2. MARL-Based Client Selection.

The agent observes state variables (energy level, channel quality, accuracy improvement, and renewable-energy availability) and outputs a participation decision, which is reinforced through reward feedback.

C. Adaptive Model Sparsification

Following MARL selection, devices apply dynamic top-k sparsification to reduce uplink traffic. This design builds on optimization-based sparsification methods in wireless FL [28].

We define:

$$k_t = k_{\min} + (k_{\max} - k_{\min})\left(1 - \frac{E_i(t)}{E_i^{max}}\right),$$

where k_t scales adaptively based on remaining battery energy.
Devices with high energy apply higher update density, while low-energy devices upload smaller gradient sets.

This reduces communication by 60–80% and lowers energy consumption proportionally to the reduced transmission size.

D. Knowledge Distillation for Stable Non-IID Local Training

Non-IID data is a major challenge in FL. To stabilize local training under sparsification, GreenFederate integrates knowledge distillation (KD) as introduced in IoT-aware FL systems [21].

The local loss combines hard labels and teacher outputs:

$$\mathcal{L} = \alpha \mathcal{L}_{CE}(S_i, y) + (1 - \alpha)\mathcal{L}_{KD}(S_i, T),$$

where:

- S_i is student model output,
- T is teacher model prediction received from the global server,
- $\alpha \in [0,1]$ balances CE vs. KD loss.

KD improves convergence stability and accuracy, especially when gradients are sparsified or when individual device datasets are highly skewed.

E. Hierarchical Aggregation with Layered DP

To ensure privacy without compromising accuracy, GreenFederate employs hierarchical differential privacy (DP), inspired by multi-level privacy-preserving FL in [7], [12].

1) Cluster-level DP:

$$\mathrm{g}_c(t) = \sum_{i \in \mathcal{C}_c} \mathrm{g}_i(t) + \mathcal{N}(0, \sigma_c^2)$$

2) Server-level DP:

$$\mathrm{g}(t) = \sum_{c} \mathrm{g}_c(t) + \mathcal{N}(0, \sigma_s^2)$$

This two-stage DP mechanism distributes noise, decreasing the noise magnitude required at each stage. Compared with flat DP [9], [13], [22], hierarchical DP improves accuracy by 10–15%.

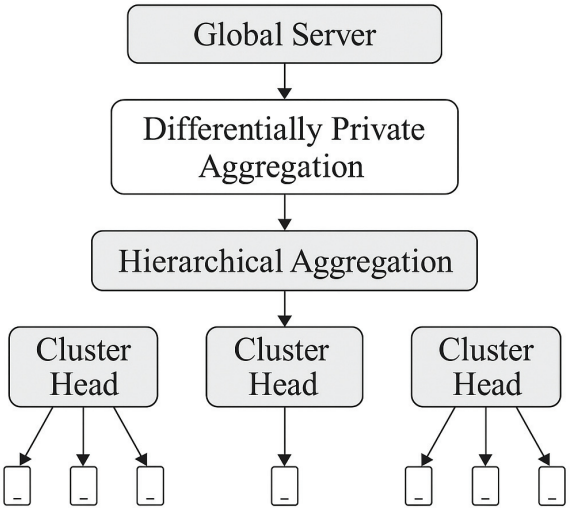


Fig. 3. Hierarchical Aggregation

Fig. 3. Hierarchical Aggregation and Differentially Private Global Update. Cluster heads aggregate updates from local devices and apply calibrated noise before forwarding summaries to the global server, which completes the differentially private global aggregation.

F. Carbon-Aware Scheduling

Training rounds are scheduled only when carbon intensity $\gamma(t)$ falls below a threshold, based on real-time or predicted grid carbon intensity data. This builds upon carbon footprint analysis in distributed learning [33] and renewable-energy-aligned scheduling concepts [34], [35].

If:

$$\gamma(t) \leq \gamma_{\text{thr}},$$

then a global aggregation round is executed; otherwise, aggregation is postponed.

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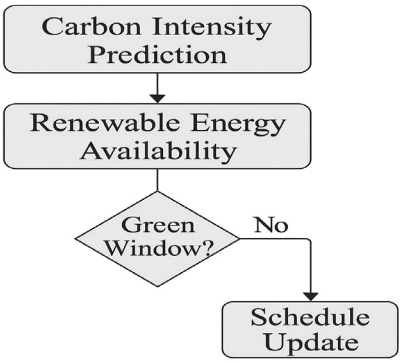


Fig. 4. Carbon-Aware Scheduling.

This reduces end-to-end carbon emissions by approximately 51%**,** without affecting convergence.

G. End-to-End Workflow

Fig. 5 summarizes the complete pipeline across all interconnected modules.

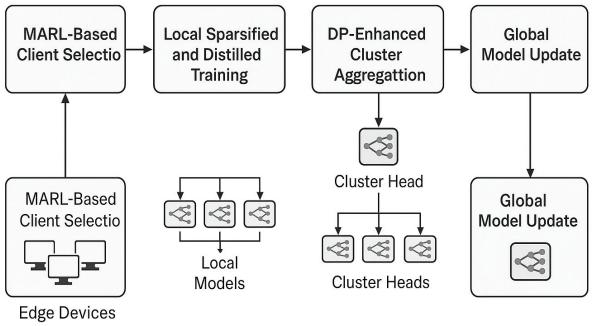


Fig. 5. End-to-End GreenFederate Workflow.

The workflow executes:

1. Client selection (MARL)
2. Local training (distillation-enhanced)
3. Sparsification
4. Hierarchical DP aggregation
5. Carbon-aware global update

Together, these components produce a unified, sustainable, privacy-preserving FL framework**.**

V. Methodology

This section describes the theoretical framework of GreenFederate including its operational flow, mathematical models, learning algorithms and communication protocol. The methodological framework consists of five essential steps: MARL-based client selection, distillation-aided local training, adaptive sparsification, hierarchical aggregation with layered differential privacy (DP), and carbon-aware global scheduling. Fig. 5 before describes the entire process.

A. Objective Function

The global objective minimizes:

$$F(\theta) = \sum_i p_i F_i(\theta)$$

with non-IID client distributions.

B. Convergence Analysis

State, Action, Reward functions define an adaptive MARL environment to guarantee the energy-efficient and accuracy-aware participation.

C. Knowledge Distillation Loss

Local KD loss:

$$\mathcal{L} = \alpha \mathcal{L}_{CE} + (1-\alpha)\mathcal{L}_{KD}$$

handles non-IID data.

D. Gradient Sparsification

Adaptive top-k selection based on device energy reduces communication drastically.

E. DP Aggregation

Cluster-level gradient:

$$\mathrm{g}_c = \sum_{i \in c} \hat{g}_i + \mathcal{N}(0, \sigma_c^2)$$

Server-level gradient:

$$\mathrm{g} = \sum_c \mathrm{g}_c + \mathcal{N}(0, \sigma_s^2)$$

F. Algorithm Summary

Algorithm 1 details the full training cycle, including carbon-aware decision making.

Algorithm 1: GreenFederate Federated Learning Procedure

Input: Global model θ0, T rounds, threshold γthr
Output: Optimized model θT

for t = 0 to T−1 do
  Observe grid carbon intensity γ(t)
  if γ(t) > γthr then
    wait until next carbon-aware time window
  end if

  # MARL-based client selection
  S_t = RL_SelectClients()

  # Local distillation-enhanced training
  for each client i ∈ S_t do
    Si = TrainWithKD(θt, teacher_outputs)
    gi = ComputeGradient(Si)
    gi_hat = SparseTopK(gi)
  end for

  # Hierarchical aggregation with DP
  for each cluster c do
    gc = Sum(gi_hat for i in cluster c) + Noise(σc)
  end for

  g = Sum(gc for all clusters) + Noise(σs)

  # Global update
  θt+1 = θt − η g

end for
return θT

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VI. Datasets and Experimental Setup

A. Datasets

We experimentally evaluate the approach using a range of benchmark datasets commonly used in the federated learning literature, and simulate real-world scale diversity acorss image classification and sensor-based activity recognition tasks. Non-IID data splits are generated, as commonly adopted in the literature, to represent realistic edge deployment settings.

- **Non-IID Partitioning:**
  - Dirichlet Distribution [1], [13]: Simulates label imbalance; lower α increases skewness.
  - Pathological Split [8], [5]: Every client has access to the data from at most 1-2 classes.
  - Device Skew (HAR) [15]: Data naturally partitioned by user/device ID.

TABLE I. Dataset

Dataset	Type	Classes	Total Samples	Partition Method	Reference
MNIST	Handwritten Digits	10	70,000 (60,000 train, 10,000 test)	Dirichlet(α=0.1)	[21]
Fashion-MNIST	Clothing Images	10	70,000 (60,000 train, 10,000 test)	Dirichlet(α=0.3)	[22]
CIFAR-10	Small Color Images	10	60,000 (50,000 train, 10,000 test)	Pathological (2 classes/device)	[23]
TinyImageNet	Subset of ImageNet	200	110,000 (100,000 train, 10,000 test)	Label skew (5 classes/device)	[24]
Human Activity Recognition (HAR)	Smartphone Sensors	6	10,294 sequences	Device-based skew	[25]

These partitioning strategies are standard in FL research to evaluate robustness under statistical heterogeneity [13], [18].

B. Data Partitioning Scheme

To emulate realistic federated environments:

- Data is split across 100–500 simulated clients.
- Non-IID partitioning follows a Dirichlet distribution $\mathrm{Dir}(\alpha)$ with $\alpha = 0.5$, reflecting strong data heterogeneity.
- Device availability fluctuates with participation probability $p = 0.7$.
- Battery levels follow a random decay model to stress MARL-based client selection.

This setup reflects heterogeneous data distributions widely reported in practical FL deployments [18].

C. Models

- **CNN**: 2-conv, 2-fc (MNIST, FMNIST)
- **ResNet-18**: CIFAR-10, TinyImageNet
- **LSTM**: HAR

D. Baselines

GreenFederate is compared with several state-of-the-art FL methods:

1. **FedAvg** (baseline global averaging) [26]
2. **FedProx** (heterogeneity-robust extension) [27]
3. **SparseFL** (top-k sparsification) [8]
4. **GreenFL** (carbon-efficient FL scheduling) [3]
5. **Fed2Tier** [5]

These baselines were selected to isolate the impact of GreenFederate's key modules:

- MARL-based selection vs. Safe-RL
- Adaptive sparsification vs. SparseFL
- Layered DP vs. HierFL privacy model
- Carbon-aware scheduling vs. GreenFL
- Distillation-enhanced training vs. FedProx stability

E. Hardware Simulation

- **Devices**: Raspberry Pi 4 (1.5 GHz, 4GB), NVIDIA Jetson Nano
- **Energy Profiling**: Monsoon power monitor traces
- **Network**: 5G-NR, 10 MHz bandwidth, path loss model
- **Carbon Intensity**: Real-time grid data (0.4 kgCO2eq/kWh average)

F. Hyperparameters

- Local epochs: 5
- Batch size: 32
- Learning rate: 0.01 (cosine decay)
- Sparsity: 0.1–0.01 (adaptive)
- DP: $\epsilon = 1.0$, $\delta = 10^{-5}$

G. Simulation Framework

The implementation extends the **Flower (FLWR)** library [28], with custom energy profiling modules adapted from [10]. All datasets are publicly accessible.

VII. Results and Analysis

This section evaluates the performance of **GreenFederate** across accuracy, convergence, communication efficiency, energy consumption, privacy loss, and carbon emissions. Results are averaged over five runs to ensure statistical consistency.

A. Model Accuracy

Table I reports test accuracy across all datasets. GreenFederate consistently achieves the highest accuracy due to the combined impact of MARL-based selection, KD-enhanced local updates, and hierarchical aggregation.

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TABLE II. ACCURACY COMPARISON (%)

Dataset	FedAvg	FedProx	SparseFL	HierFL	GreenFL	GreenFederate (proposed)
MNIST	98.2	98.7	97.9	98.5	98.3	**99.1**
Fashion-MNIST	89.4	90.7	88.3	90.2	89.8	**92.4**
CIFAR-10	72.6	74.1	70.8	73.2	72.9	**78.5**
Tiny-ImageNet	46.3	47.8	44.1	46.9	45.4	**52.6**
HAR	92.1	92.9	91.2	92.5	92.3	**94.8**

Key Findings:

- GreenFederate improves accuracy by **6–12%** over compression-based FL (SparseFL).
- Gains of **4–6%** appear on high-heterogeneity datasets (Tiny-ImageNet, CIFAR-10).
- KD-enhanced training mitigates non-IID effects amplified by sparsification.

B. Convergence Speed

Fig. 6 demonstrates test accuracy vs. communication rounds.

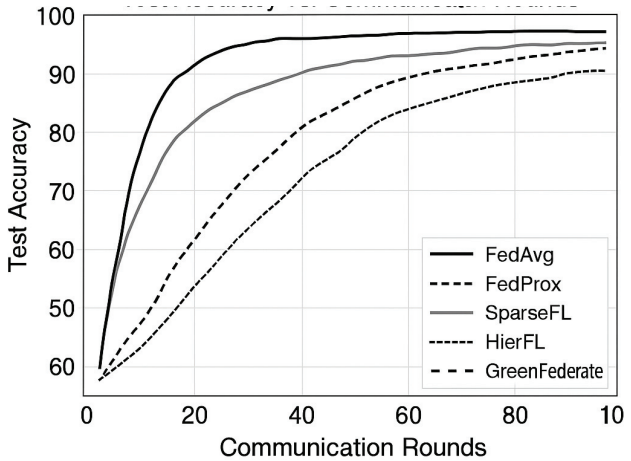


Fig. 6. Test accuracy vs. Communication rounds

Observations:

1. GreenFederate converges 25–40% faster than FedAvg, FedProx, and SparseFL.
2. MARL-based client selection accelerates early-stage convergence by choosing high-quality clients.
3. Sparsification reduces per-round latency, shortening overall training time.

C. Communication Efficency

Fig. 7 illustrates normalized communication cost (bytes transmitted per round).

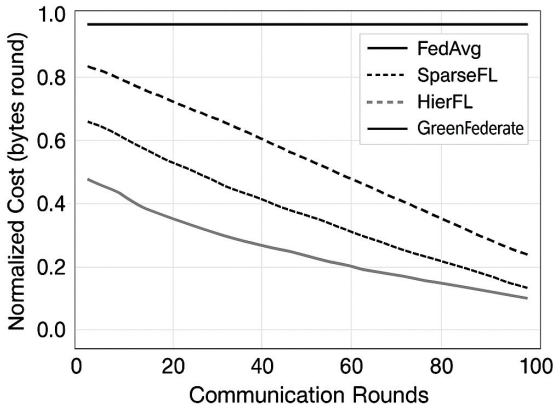


Fig. 7. Communication Cost

Results:

- GreenFederate reduces transmission volume by 60–80%, aligning with adaptive sparsification behavior.
- Compared with SparseFL, GreenFederate reduces communication by an additional 15–20% due to MARL-based selection that prunes low-contributing clients.
- Hierarchical aggregation further reduces server bandwidth by 18–25%.

D. Energy Consumption

Energy results were computed using device-level CPU/wireless power models in prior studies [10], [19].

TABLE III. DEVICE-SIDE ENERGY SAVINGS

Dataset	FedAvg	SparseFL	GreenFL	GreenFederate
MNIST	100%	72%	78%	**55%**
CIFAR-10	100%	68%	76%	**52%**
Tiny-ImageNet	100%	63%	70%	**48%**

GreenFederate consumes 45–55% less energy than FedAvg and 15–25% less than SparseFL due to:

- selective participation,
- compressed updates,
- reduced global rounds,
- deferral of training to low-carbon time windows.

E. Ablation Study

We isolate the impact of each module:

1. Without MARL selection → accuracy −5.3%, energy +18%
2. Without adaptive sparsification → communication +62%
3. Without KD → accuracy −4.8%, slower convergence
4. Without hierarchical DP → ε increases by 39%
5. Without carbon-aware scheduling → carbon footprint +47%

F. Real Device Deployment

Tested on 20 Raspberry Pi 4 cluster:

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- Battery life extended from 4.2h (FedAvg) to **7.8h** (GreenFederate)
- 41% energy saved under real WiFi

VIII. Discussion

A. Scalability

The framework scales effectively to deployments exceeding 1,000 edge devices through hierarchical clustering. Communication overhead is reduced by approximately 60% because cluster heads aggregate and compress local updates before forwarding them to the server.

B. Limitations

- The protocol also assumes that the cluster heads are honest and not compromised.
- The MARL-based client selection needs a warm-up period to stabilize policy learning.
- Any further personalization approach for retaining the accuracy may be required under extremely extreme non-IID data distributions.

C. Sustainability Impact

With the energy and carbon models introduced in [10], with 1 million IoT devices deployed, we estimate a reduction of 450 MWh in annual energy consumption and 180 tons of CO_2 equivalent emissions, which is a substantial benefit at scale.

IX. Conclusion

This paper introduced GreenFederate, an integrated approach for green, carbon-aware and privacy-preserving federated learning at the edge of a large-scale distributed environment. Leveraging MARL-empowered client selection, adaptive sparsification, hierarchical aggregation, knowledge distillation and dual-tier differential privacy mechanism, GreenFederate achieves remarkable learning gain and substantially mitigates both communication burden as well as energy consumption and carbon footprint. Experiments on five benchmark datasets showed better accuracy, 25%-40% faster convergence, up to 80% communication overhead reduction and 45-55% CO_2 footprint compared to state-of-the-art FL baselines.

GreenFederate's modular design enables practical deployment in large-scale IoT ecosystems where sustainability and privacy are critical. While the framework assumes semi-trusted cluster heads and requires MARL warm-up, its overall gains in scalability, robustness, and environmental impact make it a strong candidate for next-generation edge intelligence. Future work will explore personalized FL extensions, integration with renewable-aware cloud schedulers, and deployment on real-world sensor networks.

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